

Total No. of Questions : 10]

SEAT No. :

P3035

[Total No. of Pages : 4

[5670]-131

B.E.

REFRIGERATION & AIR CONDITIONING

(2012 Pattern) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Draw neat diagrams wherever necessary.
- 3) Use of scientific calculator, psychrometric chart is allowed.
- 4) Assume suitable data wherever necessary.
- 5) Figures to the right indicate full marks.

Q1) a) Define 'Ton of refrigeration' and prove that 1 TR = 3.517 W. [4]

b) A domestic refrigerator with a COP of 1.8 removes heat from a refrigerated space at a rate of 90 kJ/min. Determine compression power and heat rejection rate to surroundings. [6]

OR

Q2) a) State any two advantages and disadvantages of air refrigeration. [4]

b) Draw skeleton of p-h diagram and mark the property line on it. Sketch ideal vapour compression cycle on p-h diagram and mark the name of processes. [6]

Q3) a) Discuss the effect of subcooling on performance of VCC with the help of p-h diagram. [4]

b) Discuss the function of absorber in vapour absorption refrigeration system. Why cooling provision is provided for absorber? [6]

OR

P.T.O.

Q4) a) Compare vapour absorption system with vapour compression system on any four criterion. [4]

b) List six eco-friendly refrigerants and state why eco-friendly refrigerant must be used? [6]

Q5) a) Define effective temperature and discuss the effect of (i) Climate and Seasonal Variation and (ii) Density of Occupation on optimum effective temperature. [6]

b) The air at 28°C and 1 bar has a specific humidity of 0.016 kg per kg of dry air. Determine (i) partial pressure of water vapour, (ii) relative humidity, and (iii) specific enthalpy. [10]

Use sat pressure 0.03778 bar at 28°C.

Use of psychrometric chart is not allowed.

OR

Q6) a) Define specific humidity and relative humidity. [4]

b) Use standard psychrometric chart and show the basic air conditioning processes on it. [4]

c) Air is cooled from 39°C DBT and 29% RH to 24°C at the rate of 5 m³/s. Calculate the capacity of the cooling coil if the surface of the cooling coil is 20°C. Also, calculate the by-pass factor. Refer attached chart. [8]

Q7) a) Draw a schematic of summer air conditioner and explain its working in brief. [6]

b) List the indoor and outdoor components of a split air conditioner. List any two advantages of its use over other types. [6]

c) Explain the working principle of capillary tube with schematic. [6]

OR

Q8) a) Draw $p-v$ diagram for single acting reciprocating compressor and explain its working. [6]

b) Draw a schematic of Flooded evaporator and state its advantages. [6]

c) Compare unitary and central air conditioning systems on any three parameters. [6]

Q9) a) Draw an AHU and mark its components. [4]

b) What are desirable properties of duct materials? [4]

c) Explain the types of fan used in air condition system for supply and return ducts. Write any two types and their causes of pressure losses in ducts. [8]

OR

Q10) a) Prove that the equivalent diameter of rectangular duct for same air flow rate is given by [8]

$$D_{eq} = 1.265 \left[\frac{(ab)^3}{(a+b)} \right]^{1/5}$$

where a/b is aspect ratio.

b) Draw perimeter type and extended plenum duct arrangements and explain in brief. [8]

